Remotely Piloted Aircraft Systems Operations Manual

Aeria Solutions ltd

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Preamble

Aeria Solutions is a boutique environmental research firm specializing in cutting-edge RPAS (Remotely Piloted Aircraft Systems) data collection and analytics. At the intersection of technology and environmental stewardship, Aeria Solutions is dedicated to helping our clients achieve their objectives with precision and integrity. We are committed to upholding the highest standards in regulatory compliance, data analysis, and reporting, ensuring that every project we undertake is executed with excellence.

As a small, specialized firm, Aeria Solutions takes pride in our bespoke approach to each client engagement, delivering tailored solutions that reflect our deep expertise and personalized service. Our commitment to safety, encompassing the well-being of our team, the public, the environment, and infrastructure is central to our operations and success.

This Operations Manual document is a vital tool in maintaining these high standards. It provides the framework within which all Aeria Solutions personnel must operate, ensuring that every operation is carried out safely, efficiently, and in full compliance with relevant regulations. Supported by comprehensive training and supplementary materials, this Operations Manual ensures that our operators are equipped to meet and exceed our clients' expectations while safeguarding the integrity of our operations.

This document references the Canadian Aviation Regulations (CAR) as the source of aviation information.

Records and Amendments

The Amendments section is designed to document any changes or updates to the standard operating procedures. This section ensures that all personnel are informed about the latest procedures, maintaining clarity and consistency. It serves as a historical record of updates, providing accountability for changes made to the operations.

Amendment Process

Amendments to procedures within the RPAS Program can be made through two main pathways: Annual Review and On-Demand Amendments. Both methods are encouraged and welcomed, as they promote a safer, more effective program where all operators can contribute to improvements.

Annual Review Process:

- An annual review is conducted with key team members and staff.
- The purpose of this review is to assess the current status of the RPAS Program and identify areas for improvement.
- Necessary changes are proposed and follow the established amendment procedure to implement modifications.

On-Demand Amendments:

- If a field operator or group of operators identifies an issue or shortcoming in existing procedures, they can propose a solution to the Accountable Executive.
- The Accountable Executive reviews the proposal and organizes a protocol meeting to discuss any relevant details and nuances.
- With team input and approval, the proposed amendment is implemented.

Encouragement of Feedback:

- The RPAS Program highly values and encourages continuous feedback from operators.

Suggestions and proposed improvements are always welcomed to enhance safety and effectiveness for everyone involved.

Amendment Procedure

Table 1: Identification of Change

This table is used to record the basic details of each amendment made to the Operations Manual.

Steps:

1. Amendment Number: Assign a unique number to each amendment for easy identification and reference.
2. Date: Record the date on which the amendment was proposed or made.
3. Date Entered: Indicate the date when the amendment details were officially entered into the Operations Manual document.

4. Responsible Person: Identify the individual responsible for making the amendment. This person ensures the accuracy and relevance of the changes.

Identification of Change		**Date Entered**	**Person Responsible**
Amend. No.	**Date**		
Amendment 1	2024-09-16	2024-09-16	Dustin Wales
Amendment 2	2024-09-17	2024-09-17	Dustin Wales
Amendment 3	2024-09-17	2024-09-17	Dustin Wales
Amendment 4	2024-11-08	2024-11-08	Dustin Wales
Amendment 5	2024-11-11	2024-11-11	Dustin Wales
Amendment 6	2025-01-21	2025-01-21	Dustin Wales
Amendment 7	2025-01-23	2025-01-23	Dustin Wales
Amendment 8	2025-01-23	2025-01-23	Dustin Wales
Amendment 9	2025-04-30	2025-04-30	Dustin Wales

Table 2: Amendment Reference and Description

This table provides detailed information about each amendment, allowing for a comprehensive understanding of the changes.

Steps:

1. Amendment Reference: Use the unique amendment number from Table 1 as a reference.
2. Amendment Description: Provide a clear and concise description of the amendment. Include the specific section or procedure that was changed, added, or removed.

Amendment Reference	**Amendment Description**
Amendment 1	Added the requirement that all equipment have a dedicated maintenance record document which is filled accordingly.
Amendment 2	Amend the Records Checklist to utilize AirData
Amendment 3	Amend the Maintenance Program to utilize AirData
Amendment 4	Include procedure which leads to an amendment
Amendment 5	PIC most important responsibility while the RPAS is armed or in flight.
Amendment 6	Revised Site Survey for better clarity and risk assessment
Amendment 7	Add operations documentation flow to general procedures
Amendment 8	Add pilot practice requirements

Amendment 9	Revise Pilot to Pilot control hand off & Control to Control handover

Aeria Solutions

Proof of Corporation

Number: BC1132579

CERTIFICATE
OF
CHANGE OF NAME

BUSINESS CORPORATIONS ACT

I Hereby Certify that SKY PILOT UNMANNED AERIAL SOLUTIONS LTD. changed its name to AERIA SOLUTIONS LTD. on May 26, 2020 at 10:02 AM Pacific Time.

ELECTRONIC CERTIFICATE

Issued under my hand at Victoria, British Columbia
On May 26, 2020

CAROL PREST
Registrar of Companies
Province of British Columbia
Canada

Proof of Insurance

Proof of insurance is included in the documentation folder and may be distributed to anyone upon request.

Team Competencies

General Requirements for All Crew Members

- Age Requirement: All members must be at least 18 years old.
- Policy Adherence: Crew members must comply with the procedures and policies outlined in this and related documents.
- Fitness for Duty: Members must be physically and mentally fit for their roles.
- Certification: Appropriate certifications and licenses are required.

Crew Fitness

To comply with CAR 901.19, crew members must not operate an RPAS if:

- Fatigue: Crew must have a minimum of 10 hours of rest.
- Fatigue: Aeria requires that no pilot operate 4 hours consecutively without a 15-minute break.
- Fatigue: No pilot can fly for 10 consecutive hours without a mandatory 10-hour rest break.
- Mental Health: Mental well-being is critical. Any signs of stress, anxiety, or other issues must be addressed promptly. Members should seek help and not participate in operations if unfit.
- Alcohol Consumption: A strict 12-hour 'bottle-to-throttle' rule is enforced. Crew members must not operate if still feeling the effects of alcohol.
- Drug Use: Crew members must not use any substances that impair their abilities, endangering safety.

Crew Certification Requirements

For all RPAS operators the following certifications are necessary:

- Basic RPAS Operator's Certificate: Required for crew members not directly piloting RPAS.

- Advanced RPAS Operator’s Certificate: Pilots must have an advanced operator’s license. It's essential to stay updated with current regulations and certifications.
- Any applicable recency requirements as required for the certification.
    - For pilots, submit the RPAS 24-month recurrent training program (as per 901.56(1)(b)(iii) of CARs) to management.
    - For pilots, you must have access to a copy of your recency examination while operating RPAS as per CAR 901.55 and 901.64)
- First Aid Certification: All crew members must have current Emergency First Aid & CPR certification. At least one member on-site must be certified in Wilderness & Remote First Aid.
- ROC-A Certificate: Required for any crew member responsible for aeronautical radio communications.

Pilot Practice Requirements

All Pilots operating for Aeria Solutions must prove through flight logs (Aeria Tracks on AirData). Monthly flight minimums:

- 25 minutes of logged airtime whether during or not during a tasking.
- Minimum of 5 launch/lands
    - This can be accomplished with 5 flights of 5 minutes each
- No system requirements, can be flown with any drone including Sub 250g so long as the pilot's credentials are synched to Aeria AirData account.

Organization Structure

Each role carries specific responsibilities and duties that must be upheld diligently at all times. **The Accountable Executive retains overarching accountability** for all operational decisions, ensuring that every action aligns with legal requirements and corporate standards. The sequence of operations is as follows:

Accountable Executive

- Initiates the operation by confirming its feasibility from both legal and corporate standpoints.
- Reviews preparations after they are completed by the Operations Manager.
- Oversees and validates the independent work of the Maintenance Manager.

Operations Manager

- Receives the go-ahead from the Accountable Executive.
- Manages all logistics, paperwork, and necessary preparations for the operation.
- Passes operational control to the Pilot in Command (PIC) once preparations are in order.

Maintenance Manager

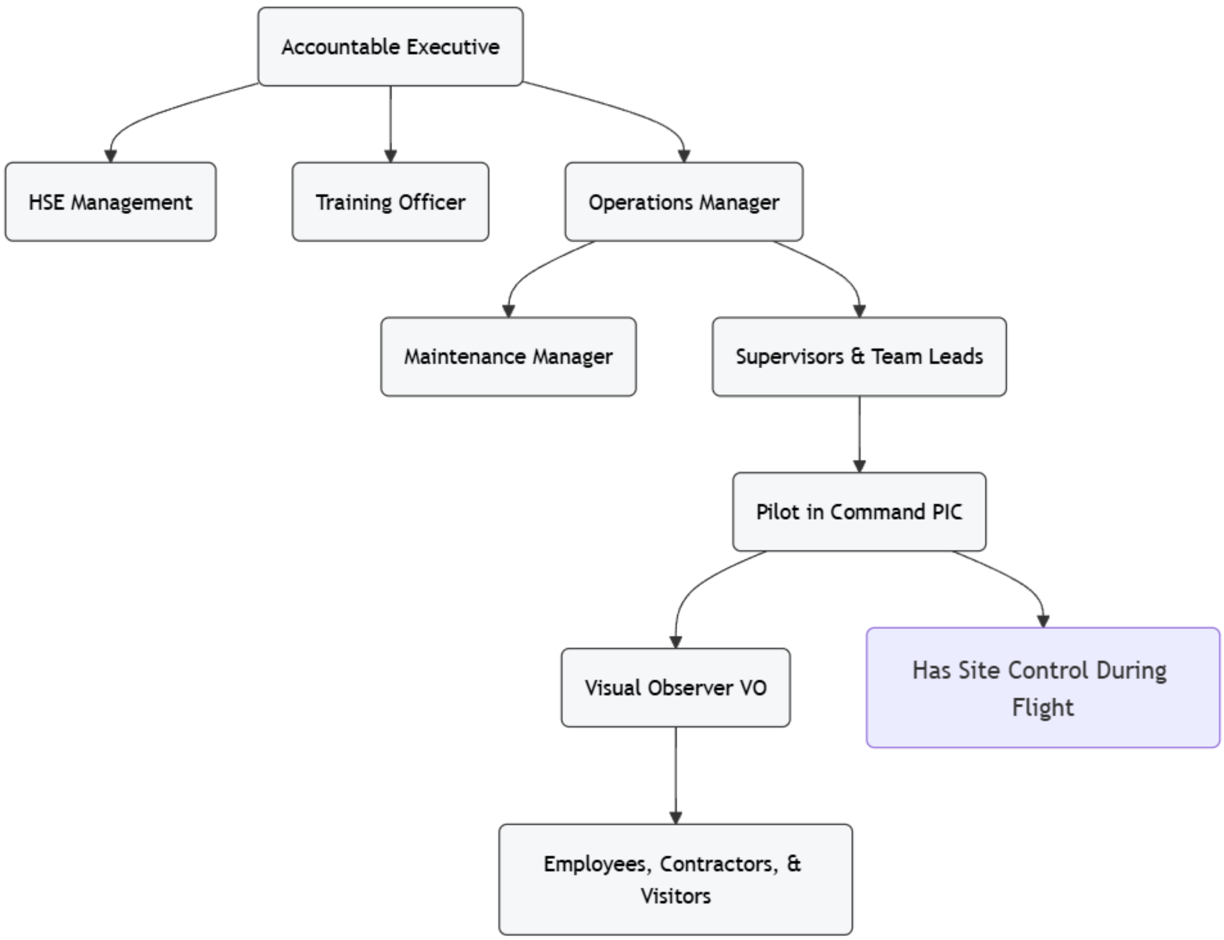
- Functions independently to manage all maintenance-related tasks.
- Work is overseen and validated by the Accountable Executive.

Pilot in Command (PIC)

- Receives control of the operation from the Operations Manager.
- Reviews all aspects of the operation, develops a plan, and conducts the mission.
- Collaborates closely with the Visual Observer (VO) to ensure safe and effective execution.

Visual Observer (VO)

- Supports the PIC by maintaining situational awareness and assisting in hazard identification and communication.



Responsibilities

Accountable Executive

The Accountable Executive is responsible for the overall operation and legal compliance of RPAS activities. This role ensures all activities comply with regulatory requirements and internal standards. Key responsibilities include:

- Enforcing operational excellence and adherence to operational plans (operational technical documentation).
- Ensuring all personnel fulfill their operational responsibilities.
- Overseeing that maintenance, storage, inventory, and tracking of equipment is executed accordingly.

- Verifying that all personnel are properly licensed and certified.
- Submitting and/or overseeing the submission of flight requests for advanced and special operations.
- Confirming operations are permitted by the appropriate regulators and authorities.
- Maintaining compliance with Special Flight Operations Certificate (SFOC) requirements.
- Confirming the accuracy of site surveys.
- Communicating with Transport Canada (TC).
- Ensuring insurance policies are current and appropriate.

Maintenance Manager

The Maintenance Manager is responsible for the upkeep and condition of all RPAS equipment. Key duties include:

- Scheduling regular maintenance and ensuring repairs are performed according to manufacturer standards
- Maintaining logs of all maintenance activities
- Verifying that all RPAS equipment complies with the latest technical and safety standards before deployment
- Ensuring adherence to hardware and software requirements as per manufacturer instructions
- Registering and ensuring the on-craft visibility of RPAS registrations
- Locking out equipment that is damaged or not maintainable
- Coordinating with manufacturers for equipment repairs or replacements
- Confirming that equipment is safe for operation

Operations Manager

- Coordinating daily operations to ensure alignment with organizational policies and safety regulations.
- Planning operations, managing flight schedules, and overseeing RPAS and personnel deployment.
- Liaising with clients and/or representatives to meet operational objectives and address concerns.
- Validating adherence to maintenance requirements.

- Validating compliance with operational planning documents and standard procedures.
- Briefing the PIC and other operation team members on the operational planning documents and parameters.
- Conducting incident/accident investigations.
- Proposing and applying amendments to operational procedures if and when they increase operational excellence and safety.

Pilot in Command

- Validating adherence to maintenance requirements
- Complying with Operations Manual and operational planning documents.
- Holding advanced operator certification and ROC-A license
- Meeting physical fitness requirements
- Staying informed on regulations and amendments
- Monitoring RPAS at all times
- Maintaining constant communication during operations
- Being trained on specific equipment
- Adhering to the training schedule

*While all of these are of the PICs responsibility it is important to note that **while the RPAS is armed and either ready for or in flight, the sole responsibility of the pilot is the safe operation of the RPAS.***

Visual Observer

The Visual Observer assists the PIC by maintaining a clear visual line of sight with the RPAS during operations. Key responsibilities include:

- Actively monitoring the airspace for potential hazards or conflicts and communicating these to the PIC
- Maintaining effective communication with the PIC
- Adhering to Operation Manual and operational planning documents
- Being aware of site complexities and any changes
- Maintaining a sterile cockpit environment for the PIC
- Reviewing incident/accident investigation.

Airworthiness and Maintenance

Overview

The Airworthiness and Maintenance section ensures that all RPAS operated by the organization meet the highest standards of operational safety and regulatory compliance. Maintenance activities are conducted according to manufacturer specifications and adhere to aviation authority regulations.

Airworthiness Standards

Aeria Solutions does not operate any RPAS which is not recognized in Canada. This includes requiring that the manufacturer has provided to the minister a safety declaration (CAR 901.76). This information can be found on the RPAS certificate of registration.

Additionally, only RPAS suitable for the operational task are permitted. For advanced or special operations (SFOC required), refer to the operations-associated planning documents for permitted RPAS information.

Maintenance Regiment

Operational Regiment

- Pre-Operation: Before field deployment, all RPAS equipment is thoroughly inspected for any nicks, dings, vulnerabilities, and battery conditions. This inspection follows the manufacturer's manual and the maintenance manager's discretion.
- Pre-Flight: Prior to each flight, a visual and tactile inspection of the RPAS and related equipment is conducted to check for any damage or issues that may have occurred during setup.
- Post-Flight: After each flight, a comprehensive visual and tactile inspection is performed to ensure no damage occurred during operation and before packing up the equipment.

Scheduled Maintenance

- Aeria utilized AirData to track maintenance requirements. AirData tracks the manufacturer's recommendation for each RPAS and includes battery and RPAS service updates based on flights and/or airtime.

- Basic Service

Maintenance is performed according to the manufacturer's program and schedule, ensuring all components are in optimal condition.

Software and Firmware Updates: All RPAS and associated operational software are regularly updated and checked for the latest versions to maintain peak performance and security.

Maintenance Recording: All maintenance activities, including updates, hardware changes or scheduled maintenance, are documented in AirData where reports can be generated upon request.

All Upcoming Drone Maintenance:

Service	Drone Name	Skip	Flights	Airtime	
Basic service is due	Default EVO2 Drone	Skip	129/20	2200/200	
Basic service is due	DJI Mini 3 Pro	Skip	130/20	414/200	
Basic service is due	DJI Mavic 3_1	Skip	119/20	456/200	
DJI Basic service is due	M300 RTK	Skip	48/450	405/9000	831/180
Basic service is due	DJI Mini 3 Pro	Skip	73/20	448/200	
DJI Basic service is due	Mavic 3 Enterprise Thermal DFO	Skip	7/450	36/9000	447/180
Basic service is due	M300 RTK	Skip	48/20	405/200	
Basic service is due	DJI Mini 3 Pro	Skip	46/20	356/200	
DJI Standard service is due	M300 RTK	Skip	48/900	405/18000	831/365
Extended service is due	Default EVO2 Drone	Skip	129/100	2200/1000	
Basic service is due	DJI Mini 3 Pro	Skip	42/20	321/200	
Extended service is due	DJI Mini 3 Pro	Skip	130/100	414/1000	
Extended service is due	DJI Mavic 3_1	Skip	119/100	456/1000	
Full service is due	Default EVO2 Drone	Skip	129/200	2200/2000	
Basic service is due soon	DJI Mini 4 Pro GKM	Skip	18/20	151/200	
DJI Premium Ext service is due soon	M300 RTK	Skip	48/2500	405/54000	831/1095
Basic service is due soon	Mavic 3 Cine GKM	Skip	15/20	151/200	
Extended service is not due yet	DJI Mini 3 Pro	Skip	73/100	448/1000	
Basic service is not due yet	Indro Robotics Mavic Pro	Skip	14/20	89/200	
Full service is not due yet	DJI Mini 3 Pro	Skip	130/200	414/2000	
Full service is not due yet	DJI Mavic 3_1	Skip	119/200	456/2000	
Extended service is not due yet	M300 RTK	Skip	48/100	405/1000	
Extended service is not due yet	DJI Mini 3 Pro	Skip	46/100	356/1000	
Extended service is not due yet	DJI Mini 3 Pro	Skip	42/100	321/1000	
Basic service is not due yet	G sparkle	Skip	8/20	45/200	
Full service is not due yet	DJI Mini 3 Pro	Skip	73/200	448/2000	
Basic service is not due yet	Mavic 3 Enterprise Thermal DFO	Skip	7/20	36/200	
Basic service is not due yet	Candrone Mavic Pro	Skip	6/20	15/200	
Basic service is not due yet	Roy-SPARK	Skip	5/20	20/200	
Full service is not due yet	M300 RTK	Skip	48/200	405/2000	

Image 1: Screenshot pulled from AirData exemplifies the maintenance tracking by AirData.

Registration Marking

Equipment Registration: All RPAS equipment is registered in compliance with CAR 901.02 ensuring regulatory adherence and traceability.

Visibility and Placement: All RPASs must carry a visible registration marking. The registration marking should be affixed to a permanent part of the RPAS's structure in a location where it does not peel off or get obscured during normal operations.

Maintenance of Markings: It is crucial to regularly inspect the visibility and legibility of the registration markings. Any faded or damaged markings must be restored to compliant standards immediately.

Personal Protective Equipment

Aeria requires that operators be equipped with appropriate PPE (Personal Protective Equipment) for the task. PPE beyond the basic requirements (outlined below) are detailed in the operational planning documents and must be accounted for when on task.

Basic Requirements:

- Eye protection (tinted if sun glare is possible)
- Gloves for handling site and equipment (not necessary during flight unless operating in cold weather)
- High viz vest

General Procedures

Operation Documentation Flow

1. Once a project has been initiated a Site Survey must be completed and include:
    a. Ground risk and control (See Job Hazard Assessments based on rating)
    b. Air Risk & Control (See Job Hazard Assessments or SORA based on rating)
    c. Wildlife Risk and Control (See Job Hazard Assessments based on rating)
    d. WorkSafe BC First Aid Assessment

Kit Preparation Flow

1. Inventory Check: Verify all necessary equipment and backups are accounted for and in good condition. Adhere to the “pre-operation” maintenance requirements.
2. Battery Check: Ensure all batteries are fully charged and properly stored.
3. Software Updates: Update all flight control and navigation software to the latest version.
4. Documentation: Pack all necessary flight documentation, including certifications, internal manuals, equipment manuals, planning documents and permissions/certificates to operate.

Team Briefing Flow

1. Overview of Operation: Discuss the objectives, expected outcomes, and any potential challenges.
2. Role Assignment: Clarify roles and responsibilities for each team member.
3. Safety Protocols: Review safety measures, emergency procedures, and communication protocols.
4. Weather and Environmental Conditions: Brief the team on current and expected weather conditions and how they may affect the operation.
5. Workload Management: Prior to the operation check in with team to make sure their comfortable with their workload and unlikely to feel overwhelmed or in need of assistance. If someone is in need of workload management, work together to assist the individual. If need be, bring in more personnel.

Site Setup Flow

1. Area Inspection: Conduct a thorough survey of the operation area for obstacles and hazards.
2. Establish Perimeter: Set up a safety perimeter around the launch and recovery area.
3. Communication Setup: Test all communication equipment and ensure clear signals.
4. Emergency Equipment: Position first aid kits and fire extinguishers for accessibility.

RPAS Setup Flow

1. Assembly: Assemble the RPAS, ensuring all components are secure and functional. Adhere to “pre-flight” maintenance requirements.
2. Calibration: Calibrate sensors and control systems.
3. System Checks: Perform a complete system check, including GPS and communications systems.
4. Battery Check: Confirm batteries are fully charged and not displaying any signs of swelling, sweating or dysfunction.
5. Final Inspection: Conduct a final visual inspection to confirm the RPAS is ready for operation.

Records Checklist

All RPAS are linked to AirData, where operations are recorded and maintained to ensure operational excellence and transparency. The Accountable Executive, Operations and Maintenance Managers and/or PIC can access these records or request them at any time. Below are two examples of AirData reports we utilize. To accomplish this, the pilot must confirm AirData is synced with the ground control prior to operating:

1. Confirm pilot credentials are logged in.
2. Confirm AirData logged in is synced with Ground Control

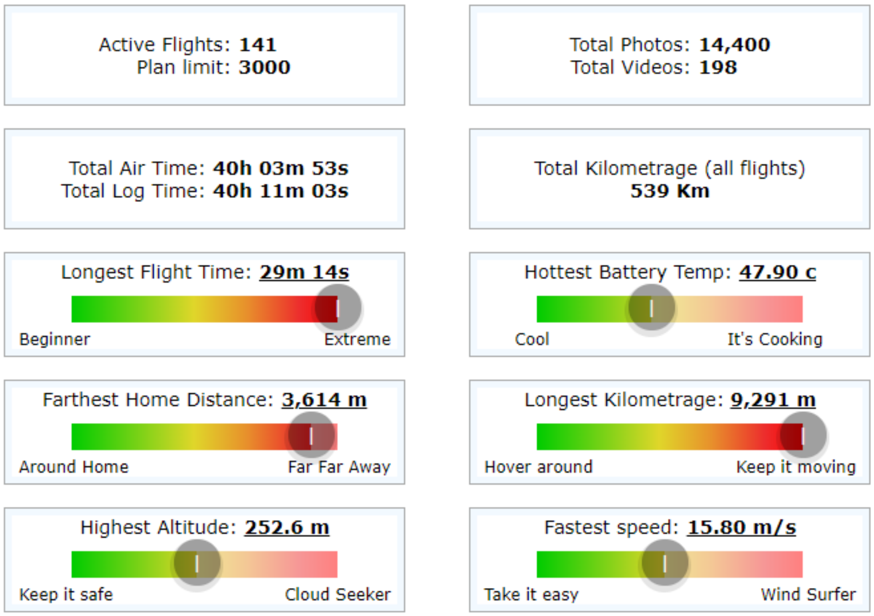


Image 2: Screenshot from AirData exemplifies a pilots log tracking

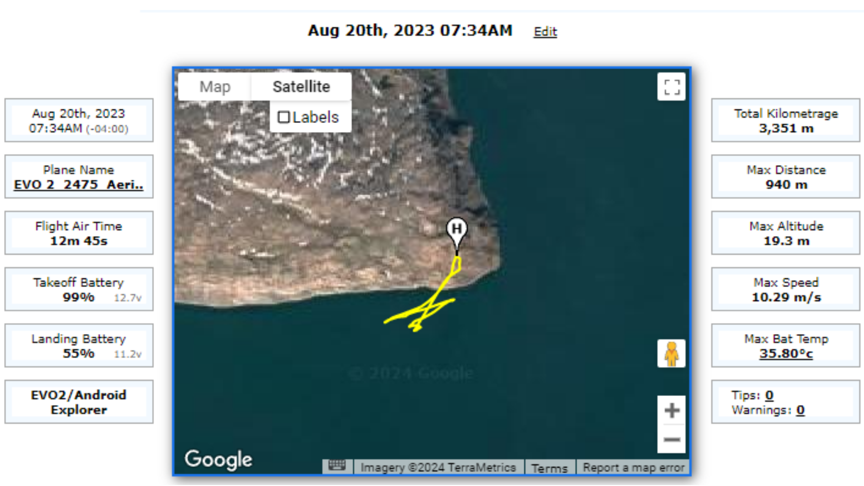


Image 3: Screenshot from AirData shows a single flight log.

Following CAR 901.48(1)(B), all records must be kept for a period of 24 months.

Operations Ready Checklist

1. Confirm all team members are present and understand their roles.
2. Verify all communication systems are functioning.
3. Ensure the RPAS and all equipment are in place and ready.

Power-Up Flow

1. Power Ground Control: Always ensure ground control is powered up first, before powering up the RPAS.
2. Energize Systems:
    a. Loudly pronounce, “RPAS ON”
    b. Power up the RPAS and all onboard systems, checking for errors.
3. Ground Control Station Setup: Establish and test the connection with the ground control station.
4. Conduct Pre-Launch Tests: confirm accurate controls of gimbal, payloads, telemetry, video and systems communication
5. Flight Settings: Confirm failsafe settings meet the parameters identified for the operation. If flying under a SFOC refer to the associated Concept of Operations (CONOPS).
6. Ready for Take-off: Inform VO you are ready for the Take-off Checklist.

Take-Off Checklist

The VO will read out the “call” and the PIC will respond accordingly. A copy of this Call and Response is included in every operation kit.

Visual Observer “Call”	Pilot in Command “Response”
“Wind & weather”	Describe wind speed and direction, and as applicable the visibility, cloud ceiling, cloud cover, mist etc. If conditions are acceptable, reply, “**Within Limits**”
“Air vehicle batteries”	Read out the charge percentage or “n” number of bars for the RPAS.
“Ground control batteries”	Read out the charge percentage or “n” number of bars. Also, if a tablet or phone is connected, read out loud the battery percentage.
“Ground control app”	Read out loud the application name, connection status, error messages, flags or required calibrations. Ensure all is ready for operation, and sufficient satellite coverage.
“Payload”	List payload types (camera, delivery box etc). check all systems are connected and nothing obstructs them.

“Failsafe”	List the failsafe mode and if applicable the RTL altitude as well as any flight distance and altitude limitations. Check critical battery level setting for failsafe.
“Take-off mode”	Should be P-GPS mode, also known asloiter. Read out the take-off mode.
“Area & Air traffic”	Note any aircraft in the vicinity, ground traffic, safety concerns, NOTAMS, obstacles and potential sources of interference (such as vessel radar, nearby people, boats, animals etc.)
“Cleared for takeoff”	Loudly pronounce “CLEAR!”

WARNING: DO NOT TAKE OFF if any of the read-outs do not meet the requirements laid out in the operation plan or the associated planning documents*.*

Take-Off Flow

1. Initiate Take-off: Engage motors and begin the takeoff sequence.
2. Monitor Telemetry: Keep a close eye on RPAS telemetry for any abnormalities.
3. Visual Confirmation: launch RPAS to a safe altitude and conduct a brief visual and audial check for concerns.
4. Attitudes and movements: conduct brief attitudes and movements to ensure accurate control of the RPAS.
5. Continue Operation: if no concerns are identified, continue with the operation.

During Flight Flow

1. Constantly monitor flight path and telemetry.
2. Adjust flight parameters as necessary based on real-time data.
3. Maintain communication with all team members.

Pre-Landing Flow

1. Confirm landing area is clear and secure.
2. Verify battery levels are sufficient for landing.
3. Check for any changes in weather that could affect landing.
4. Bring RPAS in for landing.

For certain equipment, specialized payloads are used or various techniques are applied to accommodate RPAS innovations. In these cases, refer to the manufacturer's manual or the planning documents associated with the operation.

Landing and Battery Swap Flow

1. Approach: Guide the RPAS to the designated landing area. The PIC announces the approach to ensure all crew members are aware of the landing sequence.
2. Landing: Execute a controlled landing. The PIC announces "**landing initiated**" as the RPAS begins its descent. Once landed, disarm the RPAS and allow it to power down completely. The PIC then announces to the VO that the RPAS is "**safe for approach**."
3. Battery Swap: The VO can now approach the RPAS, and exchange the depleted battery for a fully charged one. The VO will also perform a quick visual inspection of the RPAS, including the rotors and body, to check for any damage or concerns.
4. After confirming the battery is properly installed and backing away to a safe distance, the RPAS is safe to fly. The VO announces to the PIC, "**Ready for pre-flight"**
5. Pre-flight: Once confirmed safe, the PIC requests a pre-flight checklist from the VO to ensure all systems are ready for the next operation. When all checks are complete, the PIC announces "CLEAR" and prepares to continue with the Take-off Flow.

Landing and Post-Flight Flow

1. Approach: Guide the RPAS to the designated landing area. The PIC announces the approach to ensure all crew members are aware of the landing sequence.
2. Landing:
   a. Execute a controlled landing. The PIC announces "**landing initiated**" as the RPAS begins its descent.
   b. Once landed, disarm the RPAS and allow it to power down completely. The PIC then announces to the VO that the RPAS is "**safe for approach**."
   c. VO approaches, removes batteries and conducts post flight inspection.
3. Post-Flight Inspection: Conduct a thorough inspection of RPAS for any damage or wear.
4. Debrief Team: Gather the team to discuss the flight and any observations.

Pack Up Flow

1. Equipment Collection: Disassemble RPAS and collect all gear.

2. Inventory Check: Ensure all equipment is accounted for and securely packed.
3. Site Clean-Up: Clear the site of any debris or equipment.

Team Debrief Flow

1. Review the operations objectives and whether they were met.
2. Discuss any issues encountered and how they were handled.
3. Discuss any challenges had with situational awareness and how that may be avoided.
4. Gather feedback from each team member on the operation and the procedures.

Data Debrief Flow

1. Data Analysis: Review flight data to assess performance and outcomes.
2. Report Preparation: Prepare detailed operation reports highlighting successes and areas for improvement.
3. Feedback Implementation: Discuss how the insights from the data can be used to improve future operations.

Equipment Management Flow

1. Maintenance Scheduling: Schedule regular maintenance for all equipment.
2. Repair Management: Address any required repairs promptly.
3. Battery Management: Charge and ready all batteries.

Emergency Procedures

Control Station Failure

A control station failure in RPAS operations occurs when the ground-based equipment used to control the RPAS and communicate with it malfunctions or loses power. This can result in the loss of control over the RPAS, potentially causing it to operate autonomously, enter a fail-safe mode, or pose safety risks.

1. Immediate Action: Activate the RTH (Return to Home) feature if the control link is still operational.
2. Manual Recovery: If RTH is not possible, prepare for manual recovery using backup control systems.

3. Investigation and Reporting: After recovery, investigate the failure to identify and rectify the cause. Log the incident in the maintenance and operation report.

Ground Equipment Failure

A ground equipment failure in RPAS operations occurs when any support equipment on the ground, such as antennas, power supplies, or monitoring systems, malfunctions. This can impact the ability to control the RPAS, gather data, or maintain safe operations.

1. Assess: Identify the equipment failure and understand the nature and scope of the problem.
2. Mitigate: Prepare for the immediate return of the RPAS. If necessary, adjust the landing area while guiding the RPAS closer to the home position.
3. Abort Flight: Safely land the RPAS and resolve the ground control issues thoroughly before resuming flight operations.

Note: Not all equipment failures directly affect the RPAS, but they can impact overall safety. Therefore, it is crucial to promptly return the RPAS, land and deal with concerns.

RPAS Failure

An RPAS failure occurs when the RPAS itself experiences a malfunction, such as issues with its motors, sensors, communication systems, or navigation systems. This can lead to loss of control, erratic behaviour, or the inability to complete the operation safely.

1. Emergency Control: Switch to RTH or manual mode if available. Attempt to regain control of the RPAS.
2. Emergency Landing: Initiate an emergency landing procedure if the RPAS is still manageable.
3. Total Loss: If control cannot be regained and the RPAS is in total loss of control (not a fly-away), the RPAS will crash. In the event of a crash follow the “Crash Event” protocol below.
4. Post-incident Analysis: Conduct a thorough analysis to determine the cause of the failure and implement measures to prevent recurrence.

Crash Event

A crash event in RPAS operations refers to an incident where the RPAS unintentionally impacts the ground, a structure, or another object.

1. Notify: Immediately inform the VO and crew of the RPAS failure.
2. Maintain Visual Contact:
    a. Keep the RPAS in sight if possible.
    b. If operating BVLOS (Beyond Visual Line of Sight), stay attentive to any location data provided by the RPAS.
    c. Instruct the VO to scan the area for the RPAS in BVLOS operations.
3. Prepare for Retrieval:
    a. If it is safe, the operations manager should prepare the crew for RPAS retrieval.
    b. Ensure the team has a first aid kit and fire extinguisher on hand.
4. Notify Authorities: Contact the appropriate authorities immediately as outlined in the operations planning documents.

Emergency Landing

An emergency landing in RPAS operations is a procedure where the RPAS is brought down safely as quickly as possible due to an unforeseen issue or potential hazard. This is done to prevent accidents or further complications, such as equipment failure, loss of control, or adverse weather conditions. The goal is to land the RPAS in a safe location, minimizing risk to people, property, and the aircraft itself.

1. Site Selection: Quickly identify a safe landing site free from people, structures, and hazards.
2. Communication: Inform all team members and local authorities if necessary about the emergency landing.
3. Execution: Execute the emergency landing procedure as practiced in training.
4. RPAS Recovery: Secure the RPAS after landing and inspect for damage.

Fly-Away

A fly-away in RPAS operations refers to an event where the RPAS unexpectedly flies beyond the operator's control. This can occur due to a loss of communication, GPS failure, or software/hardware malfunction. In a fly-away, the RPAS may continue in an uncontrolled manner, posing risks to safety and potentially entering restricted airspace or causing damage.

1. Attempt Regain Control: Use all available means to regain control of the RPAS.
2. Activate Fail-safe: If applicable, activate the failsafe mode to bring the RPAS back or land it.
3. Notify Authorities: Immediately contact airspace control and any other relevant authorities about the flyaway. Use the provided fly-away script to ensure all necessary information is communicated clearly. Refer to the emergency contacts listed in the site survey for the operation.
4. Track and Recover: Attempt to track the flight path and recover the RPAS as soon as possible.

Emergency Fly-Away Call Script	
Introduction	"Hello, my name is [name]. I am an RPAS operator, and I am currently experiencing a fly-away situation."
RPAS Description	"The RPAS is [colour], [make], and [model], weighing [n] grams."
PIC Location	"My current location is [n] nautical miles from [aerodrome, airspace, or map feature], bearing [bearing]."
RPAS Heading	"The RPAS is flying at [n] feet above sea level, heading [heading] at [n] knots."
RPAS Potential	"The RPAS has a maximum flight time of [n] minutes, with an estimated remaining battery of [n] percent."
Maximum Travel	"I estimate its maximum remaining travel distance is [n] nautical miles."

Flight Termination

"Flight termination" refers to the deliberate and controlled ending of a flight, usually in emergency situations or when the RPAS poses a potential safety risk. This procedure is crucial to prevent uncontrolled or unpredictable behavior of the RPAS, particularly if there is a risk of entering restricted airspace, endangering people, animals, property, or violating regulations.

1. Move to Safe Location: Attempt to pilot the RPAS to a safe area away from people, animals, and infrastructure.
2. Notify Crew: Inform the flight crew of the decision to terminate the flight.

3. Reduce Altitude and Speed: Lower the RPAS to the lowest possible altitude AGL (Above Ground Level) and reduce to the slowest possible speed.
4. Shut Down RPAS: Initiate the shutdown procedure specific to the RPAS model to safely terminate the flight.
5. Safety Measures: Bring a fire extinguisher to the location and prepare for retrieval of the RPAS.

Communication Failures

A communication failure occurs when there is a loss of data exchange between the RPAS and the control station. This can involve telemetry, video feeds, or status updates. While the operator may still retain control over the drone, the loss of data can hinder situational awareness and decision-making.

1. Take Manual Control: Gain manual control of the RPAS to manage its flight.
2. Notify the Crew: Inform all team members, including the VO, of the communication failure.
3. Reduce Altitude: Lower the RPAS to a safe flying height to minimize risks.
4. Return to Landing Site: Navigate the RPAS back to the designated landing area.
5. Land and Shut Down: Safely land the RPAS and power it down.

Command and Control Link Failure

A command-and-control link failure specifically refers to the disruption of the primary link that allows the operator to send commands to the RPAS and receive control responses. This type of failure results in the operator losing direct control over the drone, potentially triggering automatic safety measures like return-to-home.

The protocol assumes communication is reestablished. If not, refer to the appropriate emergency protocol.

1. Attempt to Re-establish Communication: Toggle through different communication channels or frequencies.
2. Notify the Crew: Inform all team members, including the VO, about the communication failure.
3. Prepare for Landing: If communication cannot be re-established and the RPAS is within a safe range, prepare to land the drone as soon as possible.
4. Activate Fail-Safe Measures: Power off the control to prompt the fail-safe procedures, such as RTH or hovering in place.

5. Monitor the RPAS: Use any available telemetry data or visual observation to keep track of the drone's status and position.

Unintentional Loss of Visual Contact

Unintentional loss of visual contact in RPAS operations occurs when the operator or visual observer loses sight of the RPAS unexpectedly. This can happen due to environmental factors like fog, terrain obstructions, or the RPAS flying BVLOS. Losing visual contact can compromise the operator's ability to maintain situational awareness and ensure safe operation, increasing the risk of collision or violation of airspace regulations.

1. Immediate Notification: The Visual Observer immediately notifies the Pilot in Command about the loss of visual contact.
2. Use of Telemetry: Utilize onboard telemetry to determine the RPAS's position and guide it back into visual range.
3. Increase Altitude: If safe, increase altitude to clear possible obstructions and regain visual contact.
4. RTH Protocol: If visual contact cannot be reestablished, initiate the Return Home protocol.
5. Search: Continue searching for the RPAS until visual contact is regained.

Special Flight Operations Procedure

Special Flight Operations, as identified in Subpart 3 of CARs Part IX, allow diligent operators to conduct flights beyond the framework of basic or advanced operations. These operations adhere to the general procedures outlined in this document. Additionally, any procedural amendments or additions which may apply to the special operation will be outlined in the planning documents associated to the operations, specifically the CONOPs.

The following protocols must be followed when operating under these conditions.

Beyond Visual Line of Sight

Beyond Visual Line of Sight refers to a type of RPAS operation where the PIC controls the RPAS without maintaining direct visual contact with the aircraft. This protocol picks up from the operational moment where the RPAS goes from VLOS (Visual Line of Sight) to BVLOS.

Transition to BVLOS

Pre-Transition:

- The PIC announces the intention to enter BVLOS to their VO.
- The VO confirms readiness to take over visual monitoring of the surrounding airspace.

Eyes-In/Eyes-Out Protocol:

- PIC: "Entering BVLOS" – Signals the start of the transition.
- VO: "Ready for BVLOS" – Confirms that the VO is prepared.
- PIC: "Eyes in" – Indicates that the PIC is focusing on the control station.
- VO: "Eyes out" – Confirms the VO is maintaining visual monitoring of the airspace.

Maintain Continuous Monitoring:

- The PIC monitors the RPAS via the control station, paying close attention to telemetry and any alerts.
- The VO continues to scan the surrounding airspace for any potential hazards, particularly other aircraft.

Communication:

- Maintain regular check-ins between the PIC and VO.
- Any sighting of other aircraft by the VO must be immediately communicated to the PIC, who will take appropriate evasive action.
- The VO will request periodic read-outs from the PIC of battery and range, to aid in PIC awareness.

High Altitude

High altitude flights are any RPAS operations above 400’ AGL or 100 ft above the tallest obstruction within 200’ laterally (CAR 901.25). The protocol picks up when the RPAS is about to enter 400’ AGL.

Enter High Altitude:

- The PIC announces the intention to enter BVLOS to their VO.
- The PIC announces the intended max altitude.
- The VO confirms readiness for visual monitoring of the surrounding airspace.
- Gradually increase altitude while continuously monitoring RPAS performance.
- The VO actively monitors the airspace during ascent, scanning for potential conflicts with manned aircraft.

Continually Communicate:

- Announce when the 400’ altitude is reached.
- Airspace conditions.
- Current altitude until max altitude reached.
- Any unexpected changes in environmental conditions or RPAS behaviour.

Entering BVLOS High Altitude:

- PIC: "Entering BVLOS" – Signals the start of the transition.
- VO: "Ready for BVLOS" – Confirms that the VO is prepared.
- PIC: "Eyes in" – Indicates that the PIC is focusing on the control station.
- VO: "Eyes out" – Confirms the VO is maintaining visual monitoring of the airspace.

During high altitude operations, monitor for:

- Stability and accuracy of altitude.
- Continuous functionality of command-and-control links.
- Weather and atmospheric conditions.

Airspace Awareness:

- The VO must maintain vigilance for any manned aircraft entering the operational area.
- If manned aircraft are detected, the PIC should immediately reduce altitude or move the RPAS to a safe location.

Heavy Lift

Heavy Lift includes any RPAS that takes off over 25kg. Given the increased kinetic energy and relative risk with larger aircraft, Aeria implements a buffer zone between the RPAS to any operator.

50’ Buffer Zone:

- Launch and land must be 50’ or more from any person associated with the operation.
- The RPAS may not go directly over top of the operator or operating team when launching or landing.

Restricted Payloads

When carrying hazardous payloads it is required that plans and parameters of the operation adhere to the relative TDGR (Transportation of Dangerous Goods Regulations) & ATR (Air Transportation Regulations). Obligations defined by these regulations must be strictly adhered to and accounted for in the CONOPs.

Payload Preparation

- Compliance: Adhere strictly to WHMIS (Workplace Hazardous Materials Information System) and MSDS (Material Safety Data Sheets) standards when handling hazardous materials.
- Preparation Procedures: Develop and follow specific preparation procedures tailored to different types of hazardous materials.
- Pre-Operation Inspections: Conduct thorough pre-operation inspections to ensure the payload's integrity and safety, documenting any findings or concerns.

Payload Attachment

- Manufacturer Guidelines: Meticulously follow the manufacturer's guidelines for attaching and securing the payload.
- Post-Attachment Inspection: Implement post-attachment inspection protocols to confirm that the payload is securely attached and stable.

Personal Protective Equipment

- PPE Requirements: Equip all crew members with specialized PPE (Personal Protective Equipment) appropriate to the hazardous payload, including respiratory protection, gloves, and eye protection.
- PPE Guidance: Provide clear guidance on the types of PPE required for handling different hazardous materials.
- PPE Protocols: Implement protocols for the inspection, maintenance, and timely replacement of PPE to ensure continuous protection.

RPAS Communication Techniques

Effective communication is critical in RPAS operations to ensure safety, coordination, and regulation compliance. This section outlines the communication protocols that should be followed by all team members during the preflight, inflight, and postflight phases.

Preflight Communication

Briefing: Conduct a comprehensive preflight briefing that includes all team members. Discuss the operational objectives, roles, expected conditions, and any potential hazards. Always review any associated planning documents for all operations.

Check-ins: Establish a check-in procedure to confirm that all communications equipment is functional before the operation begins.

Emergency Communication Plan: Outline and confirm the emergency communication plan, including alternative communication methods in case of equipment failure.

Found in the planning documents *for advanced or special operations are the emergency contacts and scenarios in which to contact them. Ensure that the identified equipment necessary to make contact is included in the operations kit.*

In-flight Communication

Continuous Contact: Maintain continuous communication among the operations team. Use designated radio frequencies or digital communication channels.

Clear and Concise: Use clear and concise language. Standardize the terminology to avoid misunderstandings. Implement the use of NATO phonetic alphabet for critical communications.

Command Confirmation: Ensure that all commands and important communications are repeated back by the receiver to confirm that the message was received and understood correctly.

- For example, VO says, “Pilot, I see a manned aircraft north of us due south.” PIC replies, “Confirmed, I see a manned aircraft north of us due south, reducing altitude and hovering until safe to return home.”

Situational Awareness: Regularly update all team members on the status of the flight, including position, battery level, and any deviations from the planned route or behavior.

PIlot Handover Communication

Pre-Conditions

- Both pilots are Transport Canada-qualified for the operation class and briefed on mission, airspace, battery, and contingencies.
- Visual Observer (VO) has continuous eyes-on the RPAS (on live feed if in BVLOS)
- RPAS is in stable hover or straight-and-level flight and clear of obstacles.
- Failsafe parameters (RTH, geofence) verified; controller neck-strap fitted on OPIC.

Standard Pass-Off Sequence

1. Stabilize & Announce
   a. OPIC: “Stable—preparing hand-over.”
   b. VO: “Visual—airspace clear.”
2. Read-Back Brief (≤ 10 s)
   a. OPIC states: altitude, heading/hover, battery %, remaining flight time, any hazards.
3. Confirm Readiness
   a. IPIC: “Copy brief—ready to receive controller.”
   b. VO: “Ready—maintaining watch.”
4. Neutralize Controls
   a. OPIC centres sticks, confirms no input is required to hold position (e.g., GPS/Land mode).
   b. OPIC: “Controls neutral.”
5. Physical Pass
   a. OPIC places controller neck-strap around IPIC’s neck before releasing grip.

b. Controller passed hand-to-hand; OPIC keeps one hand lightly on until IPIC has full grasp.
c. Both pilots keep eyes on controller, VO remains eyes-out.

6. Positive Control Confirmation
    a. IPIC (thumb on left stick only): slight yaw nudge—confirms telemetry responds.
    b. IPIC: “I have control.”
    c. OPIC: verifies movement on screen, then: “You have control, I am monitor.”
    d. VO: “Control change confirmed.”
7. Resume Mission
    a. IPIC performs gentle pitch/roll test, then announces: “Control check complete—continuing.”
    b. OPIC steps back 1 m, becomes Safety Pilot/Monitor, ready to take back if requested.

Abort Triggers

- Any loss of visual or telemetry during pass → OPIC halts hand-over, re-assumes full grip, announces “Abort—maintaining control.”
- IPIC fails to confirm control within 3 s → OPIC retains PIC status.
- Unexpected aircraft/hazard called by VO → OPIC postpones hand-over until clear.

Logging Requirements

- Record UTC time of hand-over, OPIC & IPIC names, battery %, flight mode, and location in Operational Flight Log.
- Note any abnormalities in Maintenance/Incident log.

Ground-Control Station Handover (Dual-Station / Hot-Swap)

For operations where live command is transferred from one physical control station to a different station—e.g., from handheld RC to laptop GCS, or from primary to backup transmitter—while the RPAS remains airborne.

Pre-Conditions

- **Both stations** are powered, paired to the aircraft, and show identical telemetry (mode, GPS status, battery).
- **Link redundancy verified** (second station connected on approved control channel; no frequency conflict).
- **VO** maintains continuous eyes-on the RPAS.
- RPAS is **stable** (hover or straight-and-level ≥ 30 m AGL, clear of obstacles).
- **Failsafe/RTH values** confirmed equal on both stations.

Standard Hot-Swap Sequence

1. **Stabilize & Announce**
    a. **Current PIC (OPIC):** "Stable—preparing station swap."
    b. **VO:** "Visual—airspace clear."
2. **System Cross-Check (≤ 15 s)**
    a. OPIC reads aloud: altitude, heading, groundspeed, battery %, time remaining.
    b. **Incoming PIC (IPIC)** repeats values to confirm telemetry match: "Copy—telemetry matched."
3. **Arm Dual-Control Mode**
    a. OPIC enables "Dual / Instructor" or equivalent shared-link mode if required.
    b. IPIC confirms control inputs are **disabled/zeroed** on their sticks, then: "Ready to assume."
4. **Control Transfer**
    a. **OPIC:** "Transferring control in… 3-2-1—release." *(Releases command switch / toggle; keeps hands clear.)*
    b. **IPIC:** Immediately applies a small yaw or altitude input—verifies aircraft response.
    c. **IPIC:** "I have control."
    d. **OPIC:** Sees expected movement on own screen, then: "You have control—monitoring."
    e. **VO:** "Control change confirmed."
5. **Post-Swap Control Check**

a. IPIC performs gentle pitch/roll test and reads back fresh battery % and distance from home.
b. IPIC: “Control check complete—continuing mission.”
c. OPIC reverts to Safety Pilot/Monitor role, ready to resume if requested.

Abort & Failsafe Triggers

- **Station-link drop** or delayed telemetry (> 300 ms) during swap → OPIC toggles back to regain control, announces “Abort—maintaining primary control.”
- **IPIC does not obtain positive control within 5 s** → OPIC retains PIC status.
- **Any hazard call** from VO during transfer → OPIC postpones swap until clear.

Logging Requirements

- Record time, stations involved (e.g., RC → GCS laptop), battery %, and reason for swap in **Operational Flight Log**.
- Note anomalies in Maintenance/Incident log.

Post-Flight Communication

Debriefing: Conduct a postflight debriefing with all involved personnel to discuss the outcomes of the operation, any deviations from the planned procedures, and lessons learned.

Incident Reporting: If any unexpected situations or incidents occur, ensure detailed communication and reporting according to the organizational and regulatory requirements. As found in this Operation Manual under Incident and Accident Reporting.

Special Communication Techniques

Visual Signals: In cases where radio communication is not possible, use pre-agreed visual signals to communicate with the visual observer or other team members.

Silent Operations: For operations requiring silence (such as wildlife monitoring), use nonverbal communication tools like hand signals or light-based signals to coordinate actions.

Special communications for operations may occasionally be required, but they should never be necessary for the safe conduct of the operation*. If verbal communication is needed to ensure the safety of people, property, or wildlife, it must be readily available*.

Communication Equipment and Tools

Radio Systems: Utilize high-quality radio systems that are reliable over the operation's range. Ensure all team members are trained in the proper operation of the radios.

Backup Systems: Always have backup communication devices, such as additional radios, mobile phones, or satellite phones, especially in remote areas.

Digital Communication Tools: Use digital communication tools like tablets or smartphones with preinstalled apps that can provide a reliable backup for voice and data transmission.

Standard Operating Communication Protocols

Regular Checks: Perform regular checks on all communication equipment as part of the maintenance routine.

Compliance: Ensure compliance with all local regulations regarding the use of communication frequencies and equipment.

Documentation: Keep logs of all communications during each flight operation, especially those related to changes in the flight plan or emergency measures.

Detection, Avoidance, and Separation

Overview

Effective collision avoidance and separation planning are critical for the safety and efficiency of RPAS operations. This section outlines the procedures and techniques used to mitigate risks of collisions with both manned and unmanned aircraft, as well as ground obstacles.

Ground Collision Avoidance

Site Survey: Conduct a comprehensive site survey before the flight to identify and mark potential obstacles. Adhere to the findings of the associated Site Survey and record amendments made to any needed adjustments once on site.

Clearance: Ensure that the takeoff and landing zones are clear of people, vehicles, and other ground-based hazards.

Airborne Collision Avoidance

RPAS operators are required to yield the right of way to all manned aircraft, including but not limited to balloons, gliders, airships, and hang gliders, as stipulated by CAR 901.17. Adherence to this regulation is essential to prevent potential collisions, recognizing that pilots of manned aircraft may not easily detect RPAS. Operators must maintain a safe distance from other aircraft at all times, avoiding any proximity that could lead to a collision, in compliance with CAR 901.18.

Upon identifying a manned aircraft entering or approaching the operational airspace, the RPAS operator must immediately initiate evasive maneuvers to ensure no conflict arises. The preferred method of avoiding conflict is to promptly decrease altitude or otherwise maneuver the RPAS to clear the airspace safely and efficiently.

Pre-Flight Planning

Airspace Analysis: Conduct a thorough analysis of the airspace where the RPAS will operate, including known flight paths and local air traffic patterns.

NOTAMs and Air Traffic Control: Check for NOTAMs (Notice to Airmen) that might affect the operation and, if necessary, coordinate with the appropriate airspace operator to be aware of any manned aircraft operations in the vicinity.

In-Flight Procedures

Visual Scanning: From pre-flight preparations to the complete shutdown after landing, all crew members must maintain continuous and vigilant awareness of the surrounding airspace. Employ systematic scanning techniques, such as the Block Method, to enhance the ability to detect other aircraft or potential hazards effectively.

Audible Scanning: The first indication of an aircraft entering the operational airspace is often an audible cue. Upon hearing any unusual sounds, promptly identify the source and initiate appropriate safety protocols to maintain a safe operational environment.

Visual Observer: Aeria RPAS teams include a PIC and a trained visual observer to aid the PIC and monitor the surrounding airspace for other aircraft.

Electronic Systems: Utilize onboard electronic systems such as ADSB (Automatic Dependent Surveillance Broadcast) or TCAS (Traffic Collision Avoidance System), if available, to detect and avoid other aircraft.

Communication: Maintain 2-way communication channels with applicable airspace operators. These channels should only be available in the event of needed communication such as a fly-away.

Monitor Air Traffic: Using a certified ROC-A radio, monitor the appropriate airspace channel to build a mental picture of the other local traffic.

Frequency (MHz)	Usage
126.7	Uncontrolled airspace
123.2	Uncontrolled, unassigned aerodromes

Conflict Scheme

Aeria categorizes conflict scenarios as follows:

Aircraft in this scheme is used as it is the most likely scenario, however, this protocol is used for any airspace user as RPAS must give right of way to all airspace users.

- Planned Aircraft Approach: Known and expected aircraft communicating their flight path and timing before entering the operating area.
- Entered Airspace – Announced Aircraft: Aircraft that have entered the operating area with prior notice or communication.
- Unannounced Aircraft Detected – Non-Threat: Aircraft detected without prior communication but poses no immediate risk due to its trajectory and altitude.
- Unannounced Aircraft Detected – Potential Conflict: Aircraft detected without prior notice that may conflict with RPAS operations if no action is taken
- Unannounced Aircraft Detected - Critical Threat: Aircraft detected that poses an immediate and severe risk of collision, requiring urgent evasive maneuvers.

To account for these conflict scenarios, the following matrix is followed:

De-Conflict Scheme	
Scenario	**Action**
Planned Aircraft Approach	RPAS is grounded
Entered Airspace – Announced Aircraft	Restrict RPAS to below 400’ AGL and maintain at least 1 km horizontal distance from the aircraft path.
Unannounced Aircraft Detected – Non-Threat	Continue operation, maintain awareness of the aircraft in case of changing threat level.
Unannounced Aircraft Detected – Potential Conflict	Return RPAS to home or initiate hover and hold.
Unannounced Aircraft Detected - Critical Threat	IMMEDIATE DESCENT REQUIRED – DESCEND DESCEND DESCEND Protocol

Descend Protocol

Purpose: The Descend Protocol is an emergency procedure to address potential interaction with other airspace users. The following steps must be strictly followed.

When Operating Without a Visual Observer (VO):

1. Identification: The Pilot identifies a critical threat to the RPAS.
2. Manual Control: The Pilot immediately takes manual control of the RPAS.
3. Verbal Command: The Pilot loudly and clearly announces, "DESCEND, DESCEND, DESCEND."
4. Execution: The RPAS is maneuvered in a straight downward path to a safe altitude.

When Operating with a Visual Observer (VO):

1. Threat Identification: The VO identifies a critical threat and announces, "Critical Threat, DESCEND, DESCEND, DESCEND."
2. Pilot Action: The Pilot takes immediate manual control and begins descending while announcing, "DESCEND, DESCEND, DESCEND."

3. Execution: The RPAS is maneuvered in a straight downward path to a safe altitude.

General Guidelines:

- Safety Priority: The Descend Protocol aims to bring the RPAS to a safe altitude that avoids collision with other aircraft while not endangering the RPAS or anything in its descent path.
- Avoiding Terrain Impact: This protocol is not intended to crash the RPAS into the terrain. Descend to a safe altitude that minimizes risk to both the RPAS and other entities.
- Alternative Actions: If descending is not safe, maneuver the RPAS downwards and away from the aircraft's path to avoid collision.

Operational Limits

Height Restrictions: Adhere to the 400’ AGL altitude limitation (CAR 901.25(1)) or those allowed within the applicable SFOC to avoid encroaching into manned aircraft airspace.

Geofencing: Implement geofencing in the RPAS’s navigation systems to prevent it from entering restricted or unsafe zones.

SFOC: if permitted to fly special operations, maintain strict awareness and adherence to the limitations set out by the associated planning documents and SFOC issued to the specific operation.

Operational Procedures

Low Altitude Flight: When flying at low altitudes, reduce speed and increase surveillance to improve reaction time to unexpected obstacles.

Obstacle Detection: Use onboard sensors and cameras to detect and avoid ground obstacles during flight.

Horizontal and Vertical Separation: Maintain both horizontal and vertical separation from other airspace users. In the case of two RPAS pilots operating at the same time, plan your flights to be conducted at different altitudes and in locations at least 100’ horizontally from one another. Maintain constant communication regarding the whereabouts of your RPAS and its relation to the other.

Scheduling: If possible, schedule RPAS flights when it is the highest likelihood that you will be the sole operator within that airspace. This is not always feasible. When operating with another RPAS pilot on the same operation, schedule your takeoff and landings to be staggered from one another.

Avoidance Maneuvers

Manual Override: Ensure that the PIC can always take manual control of the RPAS to execute avoidance maneuvers if the automated systems fail.

Failsafe: Maintain awareness of the various failsafe functions on the RPAS, such as RTH and its many settings.

Emergency Concern: Upon identifying a potential conflict with a manned aircraft, the RPAS operator must immediately initiate evasive action. The preferred protocol is to promptly decrease altitude, hover, and monitor the situation until it is safe to return to the planned flight path. However, operators should always refer to their specific emergency protocols and use their best judgment to make the most appropriate decision based on the scenario at hand.

Minimum Weather Requirements

Overview

Weather conditions directly impact RPAS operations, and it is critical to follow the minimum weather requirements. These requirements are based on the RPAS's operational limits provided by the manufacturer, including maximum wind speed, precipitation, and temperature.

Areia enforces the following policy:

- Operational Limits: All RPAS operations must be conducted within 80% of the manufacturer's specified maximum limits.
    - Example: If the manufacturer's maximum wind speed is 10 m/s, Areia restricts operations to 8 m/s (10m/s x 0.8 = 8 m/s)

This policy provides an additional safety margin to account for unexpected weather changes and operational risks.

Compliance and Monitoring

Weather Forecasting Tools: Aeria utilizes the Environment and Climate Change Canada online forecasting tool and the Windy application for predicting weather as accurately as possible.

Real-Time Monitoring: For longer-term projects, implement real-time weather monitoring systems at the operation site to ensure immediate response to weather changes not predicted by forecasts.

Postponement and Cancellation: Operations should be postponed or cancelled if any forecasted or monitored weather conditions do not meet the minimum requirements outlined in this section.

Icing Protocol

Icing can occur at sub 0 temperatures, before and during the operations of an RPAS. Icing will affect the takeoff weight, change aerodynamic properties and prevent components from operating properly. Aeria requires that icing is not present nor likely to happen during the flight. When operating in sub 0 temperatures first refer to the RPAS specified cold weather tolerance.

Once ready to operate and the temperature is confirmed within the operating allowance, inspect the RPAS for icing forming while sitting on the launch pad. Then inspect the surrounding atmosphere. If icing conditions look to be in the flight envelope or moving towards the flight envelope, do not fly.

In review:

- Confirm RPAS cold weather tolerance
- Confirm icing is not forming
- Confirm icing is not likely throughout the flight

Accident and Incident Reporting

Overview

Proper reporting of incidents and accidents is crucial for compliance with Transport Canada regulations and for enhancing operational safety. This section delineates the protocols for documenting and reporting various incidents and accidents during RPAS operations, as per TC and NAV Canada guidelines.

Definitions

Incident: An occurrence not classified as an accident, associated with the operation of a RPAS, that could affect or has affected the safety of operations.

Accident: An event linked with the operation of a RPAS which results in serious injury, loss of life, or significant damage to property or the RPAS, occurring from the time the RPAS is activated for flight until its deactivation.

Procedures

This procedure outlines the required actions for RPAS pilots in the event of an incident or accident during operations as per CAR 901.49(1).

Immediate Cessation of Operations: The PIC must immediately cease all operations if any of the following incidents or accidents occur and immediately contact your HSE supervisor or Accountable Executive:

- Injuries to any person requiring medical attention.
- Unintended contact between the RPAS and any person.
- Unanticipated damage to the airframe, control station, payload, or command and control links that adversely affects the performance or flight characteristics of the RPAS.
- The RPAS is not kept within its horizontal boundaries or altitude limits.
- Any collision with or risk of collision with another aircraft.
- The RPAS becomes uncontrollable, experiences a flyaway, or goes missing.
- Any incident requiring a police report or resulting in a CADORS (Civil Aviation Daily Occurrence Reporting).

Incident and Accident Analysis:

- Before resuming operations, the RPAS pilot must conduct a thorough analysis to determine the cause of the incident or accident.
- Corrective actions must be implemented to mitigate the risk of recurrence.
- Use the TSB (Transportation Safety Board) Incident form below for recording the event.

Record Keeping:

- The RPAS pilot must maintain a record of the incident or accident analysis for a period of 12 months from the date the record is created.
- These records must be made available to the Minister of Transport upon request.

Reporting Requirements

The purpose of reporting and investigating incidents and accidents is to prevent recurrences. It is not to assign blame or liability.

Transport Canada

- If the incident or accident occurs under a Special Flight Operations Certificate (SFOC)—RPAS, it must be reported to TC using the RPAS Aviation Occurrence Reporting Form provided with the SFOC.

Transportation Safety Board

The following occurrences must also be reported to the TSB:

- An accident involving an RPAS weighing more than 25 kg.
- A person is killed or sustains serious injury from direct contact with any part of a small RPAS (weighing between 250 g and 25 kg).
- A collision between an RPAS of any size and a manned aircraft.

TSB toll-free: 1-800-387-3557

Direct or collect: 1-819-994-3741

Investigation Form

This form is a simple moc up that may be used. The internal form on SiteDocs is required during an RPAS Investigation.

RPAS Incident/Accident Investigation Form		
Section	**Details**	**Notes**
RPAS Information		
Type		
Model		
Registration		
Weight of RPAS		*If over 25kg immediately contact TSB*

Personnel Details		
PIC name		
Accountable exec. name		
Number of crew members		
Job title		
Client name		
Occurrence Details		
Date of occurrence		
Time of occurrence		
Geographical location		
Damage to RPAS		
Damage to environment		
Damage to property		
Injury to people		
Serious injury or death		*If yes, immediately contact TSB*
SFOC reference number		*If yes, submit an RPAS Aviation Occurrence Reporting Form provided with SFOC*
Collision with aircraft has occurred		*If yes, immediately contact TSB*
Reporter Details		
Name		
Title		
Phone number		

Investigation Process

Initial Assessment: Quickly assess the impact and scope of the incident or accident to determine the level of response required.

- If anything meets the requirements for contacting TSB, do so immediately.

Investigation Form: Using the internal investigation form, go through each section and add details accordingly. On the backside, write a detailed description of the incident.

Follow-up Actions

Corrective Actions: Implement corrective actions derived from investigation findings. Prioritize measures that address the most significant risks and compliance requirements.

Monitoring and Review: Monitor the effectiveness of corrective actions over time. Regularly review past incidents and accident reports to detect any trends or recurring issues.

Planning Documents

All Aeria operations require a site survey to be conducted. This standard ensures the operation team are aware of their surroundings in case of an emergency and ensures a high level of safety assurances.

For advanced operations (CAR 901.62) that do not fall under the SFOC categories a flght plan is required and may be submitted to NavCanada and the airspace operator for approval.

For special operations Aeria requires CONOPs to be conducted. If the operation falls under the Standard Scenarios (Appendix D AC 903-001), a SORA is not required:

- STSC-001: VLOS operation of RPA with MTOW of 25 kg to 600 kg over controlled ground areas in low-risk airspace.
- STSC-002: VLOS operation of rotary-wing RPA with MTOW of 25 kg to 150 kg over controlled ground areas in any airspace.
- STSC-003: VLOS operation of small RPA with MTOW of 250 g to 25 kg in Class G airspace above 400 ft AGL.
- STSC-004: BVLOS operation of small RPA with MTOW of 250 g to 25 kg over low-risk ground areas in low-risk airspace using Visual Observer DAA.

A SORA is required for special operations that fall outside these standard scenarios. The procedures for creating all necessary planning documents are detailed below, and additional guidance is provided in the Aeria Training Program.

	Documentation Required for Operation			
	Site Survey	**Flight Plan**	**CONOPS**	**SORA**

Basic operations	✓			
Advanced Operations	✓	✓		
Standard Scenarios (STSC)	✓		✓	
Non-STSC	✓		✓	✓

Site Survey

A comprehensive site survey is essential for safe and efficient RPAS (Remotely Piloted Aircraft System) operation. It establishes the framework for operational planning, airspace awareness, risk and hazard identification, and emergency preparedness. The following outlines the components of a site survey that align with best practices and regulatory compliance. A Site Survey form is provided in the SiteDocs login, where step by step instructions are included. This policy provides context to help operators in providing quality Site Surveys.

Operational Flight Information

This section provides key details about the RPAS operation to ensure all relevant aspects are clearly communicated. Include:

- **Plan Name**
- **Date of Site Survey**
- **Pilot Name and License**
- **Operator** (Aeria Solutions Ltd)
- **Date of the Operation**
- **RPAS Model(s)**
- **Description of Broad Operating Area**
- **Max Flight Altitude**
- **Description of Flight Profile**

- **Time of Operation (Begin and End)**
- **Additional Relevant Information**
- **Signing Authority**

Site Details

Conduct a thorough examination of the location to ensure the launch, landing, and operational areas are appropriately documented.

1. **Operational Boundaries**
   a. Describe boundaries by landmarks, including any pre-determined buffer zone.
   b. Include latitude/longitude of the center and the maximum altitude.
2. **Neighbouring Properties and Permissions**
   a. List contact information and approval status for property owners within or adjacent to the operational boundaries.
3. **Site Access**
   a. Detail how the site will be accessed (e.g., roads, trails, gates).
   b. Note any special requirements (e.g., a key or pass code).
4. **Topography and Geography**
   a. Assess the terrain to ensure it supports the planned flight profile.
5. **Communication Plan**
   a. Identify how two-way communication will be maintained (e.g., local authorities, FIC Kamloops in Class G)
6. **Predominant Weather**
   a. Describe typical weather patterns for the region and time of operation, using historical data where applicable.

Wildlife Assessment

Provide a dedicated review of potential wildlife and environmental considerations, assigning a risk value in accordance with Aeria's internal wildlife metrics. A wildneress Risk Assessment form is found in Aerias form folder, once compiled can be referenced here. It covers such details as:

1. **Species and Habitat Identification**

a. Note any protected species, environmentally sensitive areas, or migratory paths within the operational boundaries.

2. **Regulatory Constraints**
   a. Identify any local, provincial, or federal wildlife regulations that could impact the operation.
3. **Risk Value (Aeria Metrics)**
   a. A risk value is found when conducting a Wildlife Assessment (Low, Medium, High). For which, Job Hazard Assessments have been prepared and provide guidance for controls and measures to take.

Wildlife Management

Following the Wildlife Assessment, outline the measures or protocols put in place to mitigate any identified risks to or from local wildlife.

1. **Mitigation Strategies**
   a. Adjust flight times to avoid periods of peak wildlife activity, if applicable.
   b. Assign additional observers to monitor wildlife movement.
   c. Modify flight paths or altitudes to reduce disturbance of sensitive areas.
2. **Monitoring and Response**
   a. Establish protocols for real-time observation of wildlife presence during operations.
   b. Define actions to be taken if wildlife enters the operational area unexpectedly.
3. **Regulatory Compliance**
   a. Ensure all local, provincial, and federal wildlife protection regulations are followed throughout the operation.

Ground Risk

Identify and evaluate all potential ground-based hazards and exposures that could impact safe RPAS operation.

1. **Hazards Assessment**
   a. Identify natural or manmade structures (e.g., buildings, towers, power lines).
   b. Conduct both pre-arrival and upon-arrival assessments to confirm potential hazards.

2. **Population Density**
    a. Use Canadian census data to estimate population density within the operational boundary.
3. **Gatherings of People**
    a. Note any scheduled events or large gatherings within or near the operational area.
4. **Ground Risk Calculation (Aeria Metrics)**
    a. A risk value is found when assessing Ground Risk based on population and control (in accordance with Transport Canada). This is;
        i. Controlled or fewer than 5 people per square kilometer
        ii. Sparse/light or between 5 and 500 people per square kilometre
        iii. Suburban/event or between 500 and 50,000 people per square kilometer
    b. For each, Job Hazard Assessments have been prepared and provide guidance for controls and measures to take.

Ground Management

Outline measures taken to manage and mitigate identified ground risks, ensuring safe operations.

1. **Mitigation Strategies**
    a. Follow the Operations Manual or site-specific protocols to reduce hazards.
2. **Security Measures**
    a. Define how the launch/landing site is secured (e.g., signage, fencing, staff presence).
3. **Minimum Separations**
    a. Unless otherwise approved (e.g., under an SFOC), maintain 100 feet from uninformed/non-participating persons.

Airspace Details

A thorough airspace awareness survey is conducted for every operation. Tools like ForeFlight are used to identify airspaces within and surrounding the operational boundary.

Include for each identified airspace or aviation map feature:

- **Name or Aviation Map Feature**

- **Location Relative to Operational Boundary**
- **Distance and Direction**
- **Operator Information** (authority responsible)
- **Type of Airspace or Feature** (controlled, restricted, etc.)
- **Contact Information** (e.g., ATC, flight service unit)

Air Risk

Evaluate potential airborne risks and their possible impact on the RPAS operation.

1. **NOTAMs and Known Routes**
    a. List all applicable NOTAMs, known flight corridors, and typical altitudes of other airspace users
2. **Manned Aircraft Traffic**
    a. Identify nearby airports, heliports, or frequent manned aircraft activity.
3. **Air Risk Calculation (Aeria Metrics)**
    a. A risk value is found when conducting a Air Risk Evaluation (in accordance with Transport Canada) they are;
        i. Basic
        ii. Advanced
        iii. Special Flight
    b. For which, Job Hazard Assessments or requirements for SORAs have been prepared and provide guidance for controls and measures to take.

Airspace Management

Describe procedures for managing identified airspace to mitigate air risk effectively.

1. **Maintaining Airspace Awareness**
    - Outline how the RPAS operator and crew will monitor air traffic (e.g., radio, visual observations).
2. **Communication Procedures**
    - Detail how two-way communication with relevant authorities (e.g., ATC, FIC) will be established and maintained.

Emergency Preparedness

Provide an emergency response plan tailored to the site and operation, covering both RPAS malfunctions and broader incidents (e.g., medical emergencies, natural disasters).

1. **RPAS-Related Emergencies**
   - Adhere to the Emergency Procedures in the Operations Manual.
   - Contact information for a fly-away is included in the Airspace Awareness component.
2. **Key Elements of the Emergency Plan**
   - Emergency escape routes
   - Nearest medical facility and its contact details
   - Airspace operators' contact information
   - Immediate action instructions (e.g., "Call 911")
3. **Flight Suspension or Termination**
   - Ensure all team members know the process to safely suspend or terminate flight operations if necessary.

Identified PPE

List all **Personal Protective Equipment (PPE)** necessary for the specific job or site. This may include but is not limited to:

- High-visibility vests
- Protective eyewear
- Hearing protection
- Hard hats
- Steel-toed boots
- Gloves

Ensure that all crew members and relevant personnel adhere to any additional site- or project-specific PPE requirements.

First Aid Assessment

Conduct the WorkSafe BC First Aid Assessment form (found in the forms folder) for the project and implement required first aid measures according to the WorkSafeBC matrix.

Implementation

Compile the site survey findings into a comprehensive report, which is then reviewed by the operational team. Provide this report to all team members well in advance of the operation to inform planning, briefing, and eventual debriefing.

Record Keeping

Retain records of all site surveys for at least 24 months. These documents:

- May be requested by Transport Canada.
- Serve as valuable data for continuous improvement of operational and emergency response practices

Flight Plan

In the event of an operation falling within the advanced RPAs operations as outlined in CAR 901.62, a flight plan to NavCanada or the airspace operator is required. The following information must be compiled and submitted prior to operation and approval granted in written form.

- The date, time, and duration of the operation.
- The category, registration number, and physical characteristics of the aircraft.
- The vertical and horizontal boundaries of the area of operation.
- The route of the flight to access the area of operation.
- The proximity of the area of operation to manned aircraft approaches, departures, and traffic patterns.
- The means by which two-way communications with the appropriate ATC unit will be maintained.
- The name, contact information, and pilot certificate number of any pilot of the aircraft.
- The procedures and flight profiles to be followed in the case of a lost command and control link.
- The procedures to be followed in emergencies.
- The process and the time required to terminate the operation.
- Any other information required by the ANSP that is necessary for the provision of air traffic management.

Concept of Operations

Introduction

A Concept of Operations is a detailed and clearly defined description of the operational intentions for the RPAS operation. It outlines the operation's objectives, and the RPAS selected to achieve them, and provides clear evidence of the RPAS's suitability for the task. The CONOPS includes descriptions of the processes, procedures, and expected operating environment. It follows the requirements outlined in CAR 903.02 for operations conducted under an SFOC. Compliance with the CONOPS is mandatory, as required by the SFOC issuer.

The following material provides a template for creating a CONOPS for Aeria operations, detailing the structure and information required. A form is provided in SiteDocs for creating a CONOPs.

A sample CONOPS is included in the resources.

Preamble

The CONOPs introduction provides the following:

Preamble: introduces the document as a strategic guide to outlining the operational framework for a specific RPAS operation. It briefly details the operation's objectives, selected RPAS and suitability, procedures for the operation and adherence to CARs and TC guidelines. This section aligns stakeholders with the operational plan.

Background: provides context for the RPAS operation, detailing the purpose of the operation, operational environment and any relevant historical, situational or stakeholder factors.

Objectives: clearly defines the specific goals and outcomes that the operation is intended to achieve. It details the operational targets, performance indicators and other criteria of operational success, ensuring all stakeholders are aligned on the purpose and expectations of the operation.

Site Survey

Location Selection: Provide a brief description of the criteria used to select the site for the operation. Highlight why this location is suitable for the operation.

Site Characteristics: List and describe the specific characteristics that make the site eligible for the operation, particularly if there are regulatory requirements that the site must meet (e.g., STSC eligibility). Include the following:

Population Density: Specify the site's distance from areas with significant population density (e.g., minimum distance from areas with a population density greater than 25 ppl/km²).

You may have already accomplished this in a Site Survey, feel free to copy and paste the information across documents for consistency.

Controlled Ground Area: Describe the control measures in place for the site, including the geographical extent of the controlled area (flight geography plus contingency volume) and any required buffer zones.

Airspace Classification: State the classification of the airspace (e.g., Class G) and its compliance with regulatory requirements, such as distance from aerodromes, airports, or controlled airspace.

In cases where a CONOPs covers a large geographical area with multiple airspace classes and aviation features, identify the various aspects and describe the measures taken for each.

Description of RPAS

The RPAS Description section serves two key purposes: first, to articulate the capability requirements of the project through a thorough needs analysis, and second, to confirm that the selected RPAS satisfies these requirements. During the needs analysis and equipment selection process, it is essential to ensure that only RPAS that fully comply with the technical standards specified in CAR Standard 922 are considered. This includes the requirement that all RPAS must have a Safety Assurance Declaration from the manufacturer. Aeria strictly adheres to these standards to ensure operational safety and regulatory compliance.

Needs Analysis: In point form, describe the specific flight profiles and objectives that the RPAS must fulfill. These requirements are critical for determining which RPAS is capable of meeting the project's deliverables. Examples of important considerations include:

- Flying Beyond Visual Line of Sight

- Operating at high altitudes
- Carrying payloads
- Collecting water samples
- Flying for extended durations
- Capturing high-resolution videography

The above examples would be expanded upon considering the context of the operation.

RPAS Selection: Once the appropriate RPAS (or multiple RPASs) have been selected, provide a detailed description of the system, including a breakdown of the following information:

- Physical Description
- Performance Characteristics
- Flight Battery
- Command and Control Link
- Safety Features
- Pictures of the RPAS from multiple angles

Maintenance

Refer to the Maintenance schedule found in the manual for Aeria's standard maintenance procedures and identify any unique maintenance requirements specific to the selected RPAS. If the RPAS or any of its payloads require special attention, these should be detailed as procedures to be followed and included in this section.

Always review the manufactures manual and state in the CONOPs that manufacturers maintenance schedule will be adhered to.

Team Composition

Given the requirements of the operation, a thoughtful plan must be developed to determine the appropriate team composition. The Team Composition section should list each team member, their role in the operation, and the specific responsibilities associated with that role.

It is important to note that some roles can be combined if they do not require simultaneous attention. For example, the PIC and VO must be separate individuals, but the Operations Manager and PIC roles could be held by the same person, depending on the operation's needs.

Aeria's Operations Manual provide a detailed outline of roles and responsibilities for various tasks in a typical operation. However, some operations may involve unique nuances that require additional responsibilities or the creation of new roles.

In this section, carefully consider all required tasks and assign them appropriately to the responsible individuals.

Operational Procedures

All operations have a framework they must work within. In this section a clear description of these framework, including the limitations of the operation are explained.

- Policies: All Aeria operations must adhere to Aeria Operations Manual and any operation-specific documentation, such as an SFOC. Clearly state what policies must be adhered to.
- Constraints: describes the constraints and variables that if met would demand the immediate halt of flight activities. Some of these constraints are outlined in the Operations Manual, such as if the weather meets 80% of the RPAS limitations. However other aspects may be more nuanced depending on the operation.
- Airspace management: as identified prior to the operation, operators are aware of their airspace, the permission granted and what protocols are required to operate within the airspace. Outline these protocols here and strictly adhere to them.
    - *This may have been accomplished in the Site Survey, copy and paste into this document.*
- Emergency procedures: All emergency procedures for RPAS operations are outlined in the Operations Manual. For any operation-specific protocols that are to be included, detail them here.

Description of Operations

Break down the planned flights into distinct phases and describe each phase. This serves as a step-by-step guide that any team member can follow during flight operations. The description of operations must align with the operation's requirements and plans, ensuring that only necessary and planned maneuvers are executed to minimize risk. This section functions like a methods section, detailing the operational aspects of RPAS flights.

For example: An operation requiring BVLOS flights from a single point to monitor marine vessels might include the following description (presented in a table format).

Phase	Description
Launch	*Power up and launch RPAS to above head height and hover*
Orientate out	*With eyes on the subject to be flow to, orient the drone so that a direct path can be flown to the subject*
Prepare distance	*Using a total station have a VO find the distance to the subject*
Flying out	*With distance known, fly straight to the designated distance*
Spotting	*Once at the destination, pan down and find the subject*
Tracking	*Once spotted, track the subject*
Orientate in	*Stop tracking, pan the camera up and orient the RPAS toward home*
Flying in	*Fly directly home*
Approach	*Slowly approach the landing location, reducing speed and altitude until above head level and above the launch pad*
Land	*Land the RPAS*

Declaration of Public Good

For all RPAS operations, a Declaration of Public Good is required. This declaration asserts that the planned operation serves a significant public interest or provides notable benefits to the community, society, or environment.

When preparing a Declaration of Public Good, clearly articulate how the operation aligns with public safety, environmental protection, or other societal benefits. Examples of such operations might include disaster response, search and rescue operations, infrastructure inspections, environmental monitoring, or other activities where the benefits to the public outweigh potential risks.

Specific Operational Risk Assessment

This section provides guidelines on how to integrate the JARUS SORA (Specific Operations Risk Assessment) into operational planning. The JARUS SORA is a structured approach to risk assessment for RPAS operations. For detailed procedures and additional context, please refer to the official JARUS SORA documentation.

Key Steps in the JARUS SORA Process

The JARUS SORA process consists of ten sequential steps, each building on the previous one to form a comprehensive risk assessment. Below is a high-level overview:

1. Concept of Operations Description
    a. Define the scope, purpose, and specifics of your RPAS operation.
    b. Include details such as operational objectives, area of operation, flight profiles, and RPAS specifications.
2. Determination of Intrinsic GRC (Ground Risk Class): Assess the potential ground impact of your operation based on factors like population density and RPAS characteristics.
3. Mitigating GRC Through Strategic Measures: Implement safety protocols and procedures to reduce the initial GRC.
4. Determination of Initial ARC (Air Risk Class): Evaluate the airspace risks associated with your operation, considering the class and complexity of the airspace.
5. Application of Strategic Mitigations to Finalize ARC: Employ OSOs (Operational Safety objectives) to mitigate air risks, reducing the ARC.
6. TMPR (Tactical Mitigation Performance Requirements): Apply in-flight mitigations like DAA (Detect and Avoid) systems to manage and reduce risks during flight.
7. Determination of SAIL (Specific Assurance and Integrity Level): Use the final GRC and ARC to determine the required level of assurance and safety measures.
8. Identification of Operational Safety Objectives (OSO): Identify specific OSOs required for your operation based on the determined SAIL.
9. Adjacent Area and Airspace Considerations: Assess and mitigate risks associated with adjacent airspace and areas in case of a fly-away.

10. Compilation of a Comprehensive Safety Portfolio: Document all the findings, mitigations, and safety measures in a comprehensive safety portfolio to be included in your SFOC application.

Documentation

All risk assessments, mitigation measures, and operational plans must be documented and included in the operational portfolio. This documentation will be part of the submission to Transport Canada for your Special Flight Operations Certificate (SFOC).

SFOC Application Process

In Canada, conducting complex operations with RPAS necessitates obtaining SFOCs from TC RPAS Centre of Excellence. An SFOC is required under the following complex conditions as per subpart 3 of CARs Part IX:

- RPAS with a maximum take-off weight over 25 kg
- BVLOS operations
- Foreign operators or pilots
- Operation at altitudes over 400 ft AGL
- Operation of more than 5 RPAs from a single control station
- Operation at a special aviation event or advertised event
- Operations with restricted payloads
- Operations within 3 NM of a National Defence aerodrome
- Any other operation requiring an SFOC as determined by the Minister.

The application process mandates that applicants demonstrate competence and preparedness to ensure the safety and security of their operations. The following documents are required for the SFOC application:

- Completed SFOC application form.
- Concept of Operations.
- Specific Operation Risk Assessment.
- SFOC compliance checklist (typically provided upon request from TC RPAS Center of Excellence).
- All accompanying documentation.

The procedure for applying is as follows:

1. Begin by completing the application form. This document requests a quick review of:
    a. Applicant details.
    b. Types of SFOC being requested.
    c. A brief overview of the operation.
    d. Names of involved representatives.
    e. Details of the equipment used.
    f. Pilot qualifications and details.
2. Develop a detailed CONOPS and RPAS SORA for the intended operation.
    a. If the operation is a Standard Scenario, use the Standard Scenario SORA found in Appendix D of AC 903.001.
3. Prepare the necessary information for the compliance checklist. Once prepared, reference the document where the information can be found in the appropriate column.
4. Submit all documentation collectively as a single file to the Centre of RPAS Excellence.
    a. Format the email subject line as follows: "Aeria Solutions - Company File Number (930355) - Application Type."
    b. Transport Canada will respond with an ATS (Application Tracking System) number. This number serves as a reference to your specific application. In future correspondence, replace the 'Application Type' with this ATS number in the email subject line.

If the email attachment is too large to send in one go send them in separate emails, using the same subject line and appending "Email 1," "Email 2," and so forth, until all documents are successfully sent.

Equipment Testing

Overview

Regular and systematic testing of RPAS is required to ensure operational safety and reliability. This section outlines the procedures for testing RPASs when new equipment is purchased, before each operation, following significant updates, and during annual recurrence checks.

Equipment testing is a mandatory practice within Aeria and must be strictly adhered to.

Testing Schedule

Aeria has designated four scheduled testing times that must be adhered to, they are as follows:

- New: upon purchase of ‘new to Aeria’ equipment, prior to any operational use.
- Operation: Immediately prior any operation where the RPAS is intended for use.
- Maintenance: post any maintenance activity, including software and hardare changes to the RPAS.
- Annual: Every year all equipment goes through an annual maintenance and testing regimen.

Testing Procedures

- System Start-up: verify error-free startup and readiness.
- Ground Control Checks: confirm control station meets take off satisfaction
- Payload pre-flight checks: ensure the payload is secure and functioning accordingly.
- Launch: test-controlled takeoff and stable hover.
- Communication checks: validate link stability and check for interference.
- Attitude and movements: assess directional control and altitude changes.
- Mode change: ensure smooth transitions between flight modes.
- Payload in-flight checks: verify payload operation and data quality in-flight.
- Normal flight: confirm accurate navigation and overall flight performance.
- Return to Home functions: test RTH function and safe return.

Documentation and Review

Test Records:

- Document all testing procedures, results, and any anomalies observed during the tests. An RPAS test spreadsheet is provided by Aeria.
- Issue Resolution: address any issues discovered during testing promptly. Implement corrective actions and retest to confirm the resolution before clearing the RPAS for operation.
- Lockout: if the equipment does not CLEAR all procedures, the status is LOCKOUT. Tag the equipment and do not allow it to be flown operationally until the issue is solved.

Review and Update Testing Protocols: Regularly review and update testing protocols to incorporate new regulatory requirements, technological advancements, or operational changes.

RPAS Test				
Name				
Date				
RPAS ID				
Testing Schedule	New	Pre Operation	Maintenance	Annual
Procedure	CLEAR		GROUND	
Start-up				
Ground control checks				
Payload pre-flight checks				
Launch				
Communication checks				
Attitude and movements				
Mode change				
Payload inflight checks				
Normal Flight				
RTH functions				
STATUS	**CLEAR**		**LOCKOUT**	

Compliance

All operators, whether employed, partnered, contracted, or otherwise associated with Aeria, are required to adhere strictly to the policies and procedures outlined in this document. Compliance with these standards is mandatory to ensure safe, legal, and effective operations.

By signing below, you acknowledge that you have read, understood, and agree to comply with the terms and procedures detailed in this document. Your signature confirms your commitment to uphold these standards in all relevant operations.

Failure to comply with these policies may result in disciplinary action, up to and including termination of the partnership or contract, depending on the severity of the breach.

X

Name:____________________________________

Title:_____________________________________